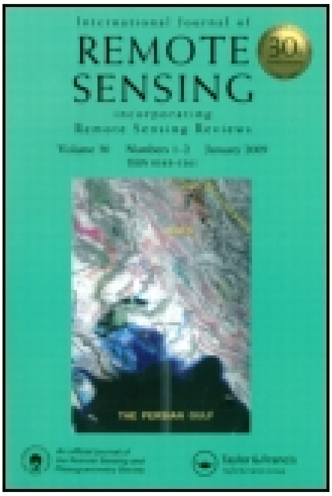

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The MERIS Case 2 water algorithm

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The paper describes the case 2 coastal water algorithm, which has been developed for the ground processor of the Medium Resolution Imaging Spectrometer (MERIS). The spectrometer is operated on board the earth observation satellite ENVISAT of the European Space Agency ESA. The algorithm derives the inherent optical properties (IOP) (1) absorption coefficient of phytoplankton pigment, (2) absorption coefficient of *gelbstoff* and total suspended matter after bleaching the phytoplankton pigment fraction and (3) the scattering coefficient of total suspended matter (TSM). The IOPs of (1) and (3) are also converted into the concentrations of chlorophyll a and TSM dry weight. The algorithm is based on a neural network (NN), which relates the bi-directional water leaving radiance reflectances with these IOPs. The network is trained with simulated reflectances. The bio-optical model used for the simulations is based on a large data set collected mainly in European waters. One special feature is the combination of a forward NN and a backward NN, which allows testing if the measured spectrum is within the scope of the training set.

The comparison with independent test data sets, in situ observations and with the case 1 water algorithm used for MERIS show a good agreement.

1. Introduction

The Medium Resolution Imaging Spectrometer MERIS has been designed mainly for ocean and coastal water remote sensing. The features that were introduced into the design of MERIS for coastal applications are the spatial resolution of 300 m together with a revisit period of one to three days (latitude dependent), nine narrow spectral bands in the range 412–708 nm including a band for measuring the sunlight stimulated fluorescence of phytoplankton, a high signal to noise ratio and high radiometric sensitivity and stability as well as a Modular Transfer Function (MTF), which is well adapted to the spatial resolution.

Since coastal waters contain a mixture of a variety of constituents with different optical properties it was necessary to develop an algorithm, which enables us to separate at least the major groups of substances, which absorb and scatter light in the visible spectrum. During the development of the ground segment it was decided by the ENVISAT project to develop two different types of ocean colour algorithms. For open ocean water, where the optical properties are determined mainly by one component, i.e. phytoplankton pigments (defined according to Morel (1988) as case 1 water), a band ratio algorithm has been developed by Morel and Antoine (2000), also to ensure consistency with other ocean colour missions (e.g. SeaWiFS, MODIS). The MERIS product, which is produced with this algorithm, is named algal_1. For areas, which cannot be described by only 1 component (case 2 waters),

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an algorithm was required, which is able to determine the three major optically relevant components, i.e. phytoplankton pigments, total suspended matter and yellow substances. It was requested that such an algorithm should be fast enough to allow mass production. This paper describes the realization of this case 2 water algorithm in its present state, which is used for reprocessing all MERIS data in 2005, and the results of performance tests. The products of this algorithm are algal_2, yellow_subs and total_susp.

There are a number of alternatives for the design of a case 2 water algorithm with the capability to decompose the IOPs into contributions by different water constituents. In the history of water colour remote sensing, Jain and Miller (1976) were among the first who used an inverse modeling technique to invert airborne reflectance spectra and to decompose the influence of different water constituents. Later Doerffer and Fischer (1994) used an optimization technique to determine water constituents from CZCS data. This inverse modeling technique includes the water and atmosphere in one model. There are now a number of similar inversion schemes available, which use so called semi-analytical models (Carder *et al.* 1999, Pozdnyakov *et al.* 2002). Further algorithms are the linear matrix inversion (Hoge and Lyon 1999, Hoogenboom *et al.*, 1998) and the linear inversion based on principle component analysis (Krawczyk 2004). However, the requirements for mass production in the ground segment are efficiency together with high accuracy. To combine these two requirements two alternatives have been analyzed: Chebychev polynomials (Schiller and Doerffer 1993) and neural network (Schiller and Doerffer 1999, Buckton *et al.* 1999). It turned out that the NN is more adequate to the water colour remote sensing problem and thus this technique was selected.

2. Design of the procedure

The NN, which is used in the ground-segment of MERIS, transforms directional water leaving radiance reflectances measured in eight spectral bands (outcome of the atmospheric correction procedure of the MERIS ground segment) and the three angles pixel by pixel with high efficiency into three inherent optical properties, i.e. (1) scattering of all particles b_tsm, (2) absorption of phytoplankton pigments a_pig and (3) absorption of gelbstoff + the bleached fraction of suspended matter a_gelb. The IOPs b_tsm and a_pig are then converted by simple equations into concentrations of the water constituents suspended matter and phytoplankton, while *gelbstoff* remains in absorption units. This separation has the advantage that the IOPs can also be used directly or converted into concentrations by other coefficients determined by the user. This advantage of deriving IOPs is also discussed in Lee *et al.* (2002).

Additionally the NN checks if its input is in the domain which was covered during the training of the NN.

The directional water leaving radiance reflectance RLw$(\theta_v\phi_v)$ associated with the water leaving radiance $L_w(\theta\phi)$ and the downwelling irradiance above the sea surface E_d is defined to be:

$$\mathrm{RLw}(\theta_v\phi_v) = L_w(\theta_v\phi_v)/E_d(\theta_s) \tag{1}$$

where θ_v and ϕ_v are the zenith and azimuth observation angles respectively. E_d depends on the solar zenith angle, θ_s, for the pixel under examination. For convenience, we will denote the wavelength dependency in the following chapters only where necessary.

Since simultaneous measurements of concentrations and water leaving radiance reflectance spectra are rare and, thus, do not cover the data space with sufficient density, the construction of the NN is based on a large table (550K entries) of simulated data generated by our forward model which was built from the HYDROLIGHT radiative transfer code (Mobley 1994) plus a bio-optical model relating scattering and absorption coefficients to concentrations.

This bio-optical model is based on a large data set of measurements of inherent optical properties and describes the variability of the three major components of coastal waters, i.e. scattering by all particles, absorption by phytoplankton pigments and absorption by humic organic matter (*gelbstoff*). The optical data are mainly from cruises in the North Sea, partly in the Baltic Sea, Mediterranean Sea and North Atlantic (see §4).

For given concentrations/IOPs of water constituents the forward model calculates the angular distribution of water leaving radiance in eight visible MERIS bands. These angular distributions are sampled in the appropriate angle ranges to derive the entries of the training/test tables for building the NN: three concentrations, three angles and eight water leaving radiance reflectances. The concentrations of the water constituents are randomly sampled from an exponential distribution in order to disentangle small concentration differences in regions of small concentrations. In order to get roughly constant relative concentration errors the logarithm of the IOP's was used as NN output.

To avoid extrapolation from the training set the NN has to learn the variability of the inherent optical properties (IOP) of the water constituents as well as the errors in reflectances caused by the instrument and the atmospheric correction. Thus, the natural variability of the IOPs was built into the forward model by sampling the parameters describing the spectral dependence of the IOPs from their measured distributions.

Two NN's are trained with this table: (1) *invNN* to emulate the inverse model $\mathbf{c}=F^{-1}{}_g(\mathbf{r})$ from reflectances $\mathbf{r}$ and geometry information g, and (2) *forwNN* to emulate the forward model $\mathbf{r}=F_g(\mathbf{c})$ deriving reflectances $\mathbf{r}$ from concentrations $\mathbf{c}$ and geometry information g. The two NN's are combined to give a new NN (figure 1), which first uses the *invNN* part to obtain an estimate of the concentrations $\mathbf{c}$. These together with the geometry information g are fed into the *forwNN*. If not only the measurement of the reflectances but also the model is perfect, the identity $\mathbf{r}=F_g(F^{-1}{}_g(\mathbf{r}))$ should hold. Therefore the reflectances $\mathbf{r}'$ returned by the *forwNN* are compared with the measured ones. Large deviations signal a violation of the necessary condition for a successful inversion; corresponding pixels are then flagged (Doerffer and Schiller 2000).

NN's are known to be good interpolators and bad in extrapolations. Therefore for the training of *invNN* an error model was applied to the reflectances obtained in the forward model run (until now only the pre-launch error model is available from MERIS simulations and tests).

The inverse NN presently in use is made of four hidden layers with 45, 16, 12 and 8 neurons, respectively. Inputs are the logarithms of the reflectances and output the logarithms of the three IOPs.

The program to train the NN is described in (Schiller 2000).

In summary the algorithm development consists of the following steps:

- Set up of a bio-optical model based on measurements of IOPs and concentrations including the natural variability of IOPs.

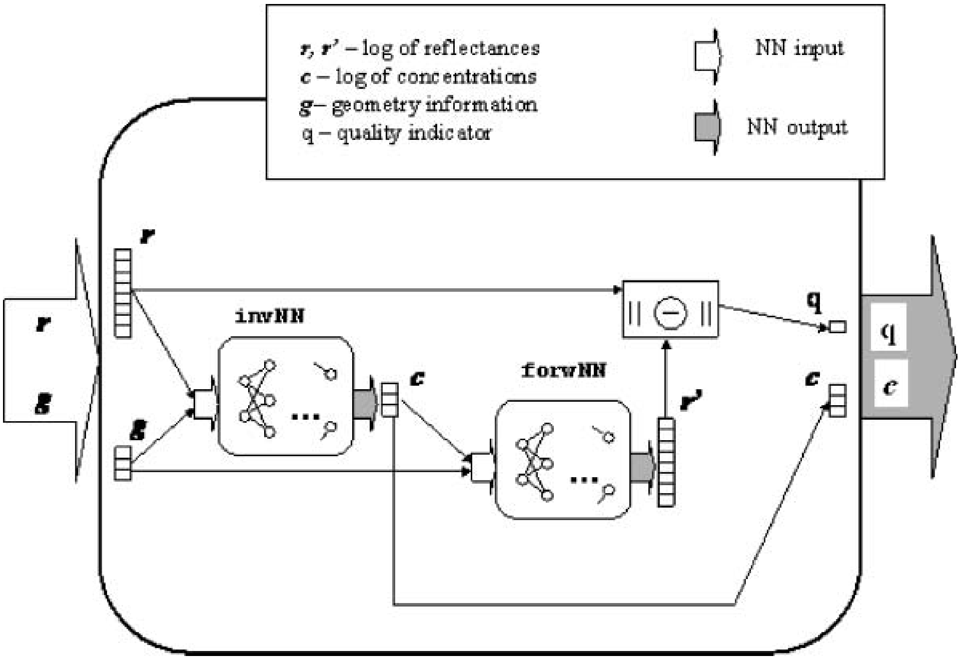

Figure 1. A combination of NN's is used in the MERIS ground segment for the retrieval of water constituent concentrations **c** from remotely sensed reflectances **r** and geometry information *g*. The **c** obtained from the *invNN* emulating the inverse model is used together with the geometry information *g* as input to the *forwNN* which calculates the corresponding reflectances **r**′ which are compared with the measured ones to give a quality measure *q* of the retrieval.

- Definition of the range of concentrations and further boundary conditions (s. bio-optical model).
- Estimation of expected errors of RLw's based on instrumental errors and uncertainties of atmospheric correction.
- Simulation of RLw spectra.
- Training of the neural network.
- Test of the network.
- Use of the network within the MERIS processor.
- Validation of the algorithm and the products.

2.1 *Special features: chi2 out of training range flag and cut off reflectance*

Two special features were introduced into the algorithm. One is the combination of the inverse NN with the forward NN in order to check on a pixel by pixel basis whether the input reflectance spectrum is within the range of the simulated set of reflectances used for training of the NN (s. description above). The degree of agreement is determined by q=sum((logn(RLw_measured)-logn(RLw_fNN))**2). The threshold to trigger the out of scope flag is presently set to a value of $q=4.0$.

The other feature is the cut-off reflectance. The RLw, which are retrieved from the top of atmosphere radiances by the atmospheric correction procedure can be significantly inaccurate or even negative at low reflectances, e.g. in the red part of the spectrum in case 1 waters due to the high absorption of pure water or in the blue part in case of high concentrations of yellow substances. Since we use the logarithm of the reflectances even small absolute errors of these bands can degrade the performance of the NN significantly. In order to cope with such faulty input the NN was trained in such a way that the reflectance was set to log(RLw) of $-6.9\,\mathrm{sr}^{-1}$ whenever the reflectance is below this minimum. Accordingly the measured reflectances after atmospheric correction were treated in the same way. However a minimum of 3 bands with reflectances above this threshold is required to run the NN algorithm. This cut off feature improved the results significantly, because low reflectances data with a relative high noise or negative reflectances are excluded.

3. The bio-optical model

Water constituents comprise a large number of different substances, which include mineralic dissolved and particulate compounds, a large variety of organic macromolecules, living organisms such as phytoplankton, zooplankton, bacteria etc., and their debris and excrements. All of these water constituents have different optical properties concerning scattering and absorption and partly fluorescence.

For the purpose of optical remote sensing this diversity of substances has to be grouped into a small number of classes each of which includes constituents with similar optical properties and/or correlated concentrations. For the majority of the world ocean areas it is sufficient to comprise all substances into one group using phytoplankton chlorophyll *a* as a proxy for its concentration. This type of water is called case I water. The concentration of this class of substances – in terms of chlorophyll *a* – can be derived from the blue to green shift of the water colour. In many coastal waters one needs more than one class of substances to describe the variability of water colour. By tradition and experience three classes are defined: (1) phytoplankton pigment with chlorophyll *a* as a proxy, (2) the dry weight of all particles (total suspended matter, TSM) and (3) the absorption caused by the "dissolved" fraction of all water constituents (*gelbstoff*).

However, each of the three groups of substances is variable with respect to their composition and thus their chemical, physical and in particular optical properties. Any remote sensing system and retrieval algorithm has to take this variability into account.

Due to this fact it was decided for MERIS not to relate the output of the NN algorithm to the concentrations of these groups of substances but to three IOPs, which can be related to these substances.

Due to the restrictions of the MERIS processor, only three components could be defined, although others are possible. These are (1) phytoplankton pigment absorption a_pig, (2) absorption of all other substances a_gbp (dissolved *gelbstoff* a_gelb and bleached particulate matter a_bp) and (3) scattering of all particles b_tsm with the assumption that the scattering component is not absorbing, i.e. that the absorption, which modifies the volume scattering of absorbing particles, is covered by the absorption a_pig and a_bp. All three optical components are defined for the wavelength of MERIS band 2 (442 nm). Furthermore, the optical component b_tsm is represented by two types of scattering particles. One is a standard particle with a wavelength exponent of 0.4 and another white scatterer with a wavelength exponent of 0.0. This white scatterer was introduced to get a representation for coccolithophorides.

An overview of all components with their properties is given in table 1.

The components are defined according to the separation and measurement procedures (figure 2). One critical issue is the separation of the absorption of

Table 1. Variability of the optical properties and range used for the simulation of water leaving radiance reflectance spectra.

Component/property	Value range
Gelbstoff absorption wavelength exponent	0.014 + −0.002
Bleached particle absorption wavelength exponent	0.008 + −0.005
Particle scattering wavelength exponent	0.4 + −0.2
White particle scattering wavelength exponent	0.0
Phytoplankton pigment absorption	random selection from >200 absorption spectra, normalized at 442 nm (MERIS band 2)
Gelbstoff absorption a_gelb at 442 nm	0.005–5.0 m^{-1}
Particle scattering bp at 442 nm	0.005–30.0 m^{-1}
White particle scattering bpw at 442 nm	0.005–30.0 m^{-1}
Phytoplankton pigment absorption a_pig at 442 nm	0.001–2.0 m^{-1}
Minimum particle scattering at 442 nm	0.25a_pig(442nm)
Bleached particle absorption	0.1bp(442) + ran_gauss0.03bp(442)
Sun zenith angle	0–80°
Viewing zenith angle	0–50°
Difference between sun and viewing azimuth angle	0–180°

phytoplankton pigments and the absorption of other substances, which are part of suspended matter. This separation is performed by oxydation of the TSM using NaHCl. The difference in the absorption spectrum before and after this treatment is assumed to be the absorption of all phytoplantkon pigments. Another issue is the separation between dissolved yellow substance and the absorption of TSM, which is determined by the type and pore size of the filter used for separation, which traditionally was 0.45 μm and nowadays is 0.2 μm.

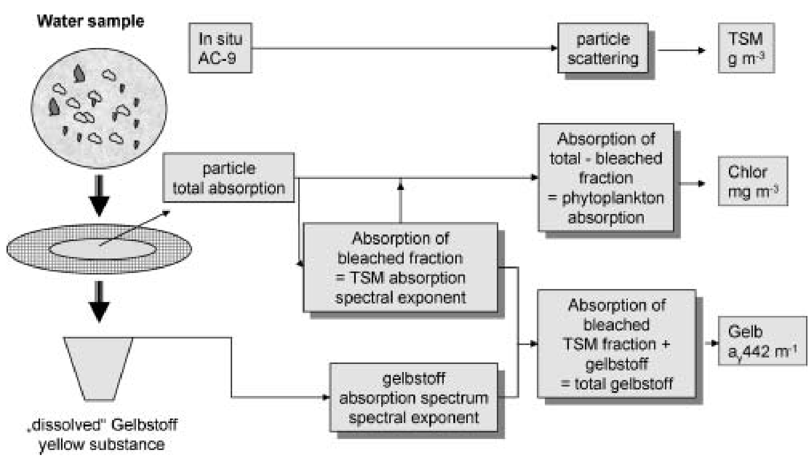

Figure 2. Scheme of the bio-optical model: The water sample is divided by filtration into a "dissolved" fraction and particulate fraction. The particulate fraction is bleached. The difference between the total particulate absorption and the absorption after bleaching is defined as the phytoplankton pigment absorption. Its value at 442 nm is a_pig. The absorption of the bleached fraction is added to the absorption of the "dissolved" fraction and provides the gelbstoff absorption at 442 nm. The scattering component is measured with the AC-9 instrument. It provides the total scattering of all particles at 442 nm.

The quantitative relationship between the concentration of a bio-geochemical component and its optical proxy is described by a conversion factor or equation, which – for MERIS – can be adapted to local conditions or exchanged by the user based on his experience.

Phytoplankton pigment absorption a_pig is related to chlorophyll *a* concentration (in μg/l), the scattering of non absorbing particles b_tsm is related to the dry weight of total suspended matter (TSM, in mg/l) and the sum of the absorption of gelbstoff a_gelb and of the bleached particulate matter, a_bp, is related to the *gelbstoff* absorption a_gelb (in 1/m). The mean conversion factors and equations are based on the results of the projects COASTLOOC and COLORS/MAPP, REVAMP, MAVT, for details s. below.

All measurements of optical properties have been carried out according to the protocols set up by ESA MAVT, which in turn is based on the protocols set up for the COLORS project and the SeaWiFS project.

3.1 *Pure water*

The absorption of pure water is based on the measurements of Pope and Fry (1997), the scattering of pure water on the formulation of Morel (1974).

3.2 *Absorption by phytoplankton pigments*

The variability of phytoplankton pigments is represented by a series of 221 spectra (figure 3) of which one is selected randomly for each simulation of the reflectance spectrum. This set of spectra has been compiled from measurements in the Southern North, in the Skagerrak and the North Sea off Norway in the period 2000–2003. All measurements have been performed according to the same protocol using the filter pad method (see Soerensen, this issue).

3.3 *Absorption by yellow substance*

Yellow substance a_gelb is defined here as the absorption at 442 nm of all water constituents which pass a Millipore filter with a pore size of 0.2 μm. The wavelength

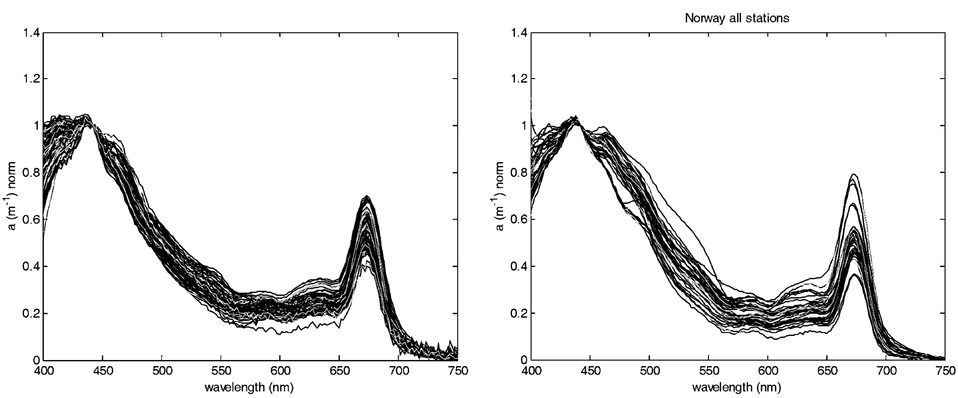

Figure 3. Absorption spectra of phytoplankton pigments (difference between non bleached and bleached fraction of total suspended matter), normalized at 442 nm, i.e. for MERIS spectral band 2. The left panel shows spectra of the Southern North Sea. The right panel shows spectra of Norwegian coastal waters including the Skagerrak (courtesy of K. Soerensen, NIVA, Norway).

exponent of the quasi exponential decrease of absorption with increasing wavelength was determined from the wavelength range 400–550 nm using a non-linear fit method. No offset substraction at 750 nm was performed. To determine the mean and standard deviation of the wavelength exponent, measurements were compiled from the Southern North Sea, the Skagerrak and the coastal waters of western Norway.

3.4 *Absorption by bleached particles*

Particles are defined as all substances, which remain on a glassfibre filter (Type Whatman GFF) after filtration. The absorption properties are determined after deterioration of the phytoplankton pigments using NaHCl. These absorption spectra look very similar to those of yellow substance but with a lower wavelength exponent. This exponent was determined in the same way as that of yellow substance.

3.5 *Scattering by all particles*

The total volume scattering *b* [m-1] of all particles has been measured in situ at 9 wavelength with an AC-9 instrument, which determines absorption and beam attenuation. This instrument was attached together with other instruments to a rosette water sampling frame, so that water samples could be taken simultaneously with the measurement. The spectral exponent was then determined from all wavelengths using a non linear fit. Furthermore, the results were checked against

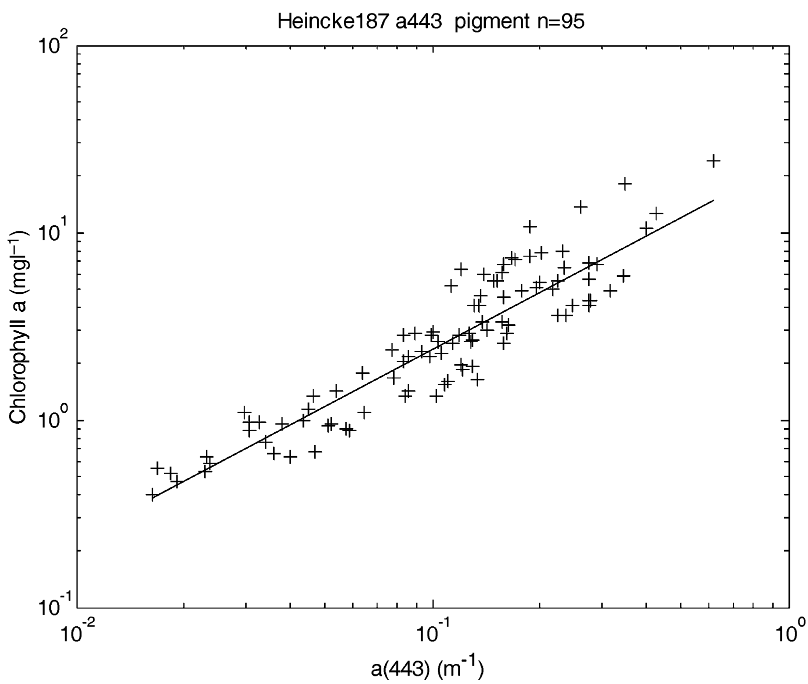

Figure 4. Relationship between pigment absorption and chlorophyll a concentration for samples of the Southern North Sea, conversion equation is algal_2=21a_pig^1.04.

data of the COASTLOOC project (Babin *et al.* 2003a), which covered coastal and off shore areas of the Mediterranean, the Atlantic, North and Baltic Sea, and compared to results of Siegel *et al.* (2005) for the Baltic area.

In addition a white scatterer (spectral exponent of 0) was introduced to include the effects of coccolithophorides, which often occurs in the North Sea and Keltic Sea during summer months. However, due to the restrictions of the MERIS products format, the white scatterer is not a separate variable, only the optical effect was introduced so that the NN learned that this kind of material may be present in the water.

The scattering phase function of Petzold as described in (Mobley 1994, table 3.10) was used for standard TSM and white particles.

The concentrations of all particles are determined as the dry weight per volume of water.

3.6 *Range and co-variations*

According to the data from North Sea Water and the concentration ranges in case 1 water (ATBD Morel and Antoine 2000) the range for pigment and gelbstoff absorption (a_pig and a_gelb) as well as particle and white particle scattering b_tsm (442 nm) was set as listed in table 1.

Although all variables should vary as independently as possible within the given ranges, we have introduced some restrictions concerning the minimum scattering associated with the absorption coefficients of phytoplankton pigments and bleached particles in order not produce unrealistic low reflectances.

These mimima were determined from the covariations between absorption and scattering.

The following relationships were determined:

$$\mathrm{b\_tsm\_min} = 0.25 * \mathrm{a\_pig}(442\mathrm{nm})$$

The co-variation between the absorption cofficient of bleached particles and the scattering coefficient of total suspended matter scattering was determined as:

$$\mathrm{a\_tsm\_bleached}(442\ \mathrm{nm}) = 0.1 * \mathrm{b\_tsm}(442) + \mathrm{ran\_gauss} * 0.03 * \mathrm{b\_tsm}(442)$$

where ran_gauss is a random number of a N(0,1) Gaussian probability distribution.

Note that the coupling excludes some extreme combinations, which however might occur in nature.

3.7 *Environmental conditions*

The environment as well as further optical properties were defined in the following way: infinite deep water (no bottom effect), vertical homogenous distribution of all water constituents, rough sea surface according to a wind speed of $3\,\mathrm{ms}^{-1}$.

The downwelling radiance distribution above water has been simulated with Hydrolight, which was adapted for this purpose to simulate an atmosphere with 50 layers for 17 solar zenith angels ranging from 0 to 80 degree.

Furthermore no inelastic scattering (fluorescence or Raman scattering) as well as no polarisation effects have been considered in the simulation.

3.8 *Conversion from IOP to concentrations*

Although the primary output of the NN algorithm are the logarithms of the three IOPs, i. e. phytoplankton pigment absorption, gelbstoff + bleached particle absorption and total suspended matter scattering, it was decided by ESA to convert the scattering coefficient into total suspended matter dry weight concentration in mg/l and the phytoplankton pigment absorption into chlorophyll *a* concentration. However, this conversion should be reversible to enable a user to go back to the IOPs from the concentrations and then may apply his own conversion factors. Thus, only single relationships between one IOP and one concentration were allowed. As for the optical properties, these relationships can be rather variable with respect to geographical regions and seasons and no global data base exists concerning coastal waters. Thus, we have based the correlations on our own measurements in the North Sea and compared the results with data of Norwegian waters provided by K. Soerensen (this issue). Furthermore, we compared the pigment absorption with the chlorophyll *a* concentrations, as derived with the case 1 water algorithm of MERIS (algal_1 product, Morel and Antoine (2000)). This reflectance ratio algorithm is based on a nearly global open ocean data set and has been developed totally independent from the algal_2 neural network algorithm. However, these comparisons are indicators; a significance test in a strict statistical sense could not be performed, because the properties of the data sets are too different.

Based on the data set of our cruises in the Southern North Sea (figure 4) we found the following relationship for phytoplankton pigments:

$$\text{chl. conc}[\mu\text{g/l}] = 21\text{a\_pig\^{}1.04}$$

This relationship was confirmed by Sorensen *et al.* (this issue) from measurements in the Skagerrak and northern North Sea along the Norwegian coast.

$$\text{Chl2.hplc} = 21{,}848 \times \text{a\_pig\^{}1{,}044}$$

As mentioned above, the relationship between a_pig and the chlorophyll concentration is also valid for algal_1:

$$\text{Algal\_1} = 21\text{a\_pig\^{}1.04}$$

but only when case 2 water pixels are excluded from this comparison (figure 5). This exclusion was done "by hand" to separate the two clusters. As it turned out, the upper cloud of figure 5 are mainly pixels from geographical regions with case 2 water. The regression line was then determined for the lower cloud.

For the conversion of b_tsm into concentration of TSM dry weight the following factor was derived from our measurements (figure 6), which is in agreement with the factor determined from the Coastlooc data set (Babin 2003a).

$$\text{TSM} = 1.73\text{b\_tsm}$$

4. Performance of the algorithm

A rigorous validation of the algorithm and the corresponding products by using a large base of match-up data of different case 2 water sites was not possible, because this data base does not exist. Reasons are e.g. the limited number of validation cruises with appropriate measurements and under appropriate weather conditions,

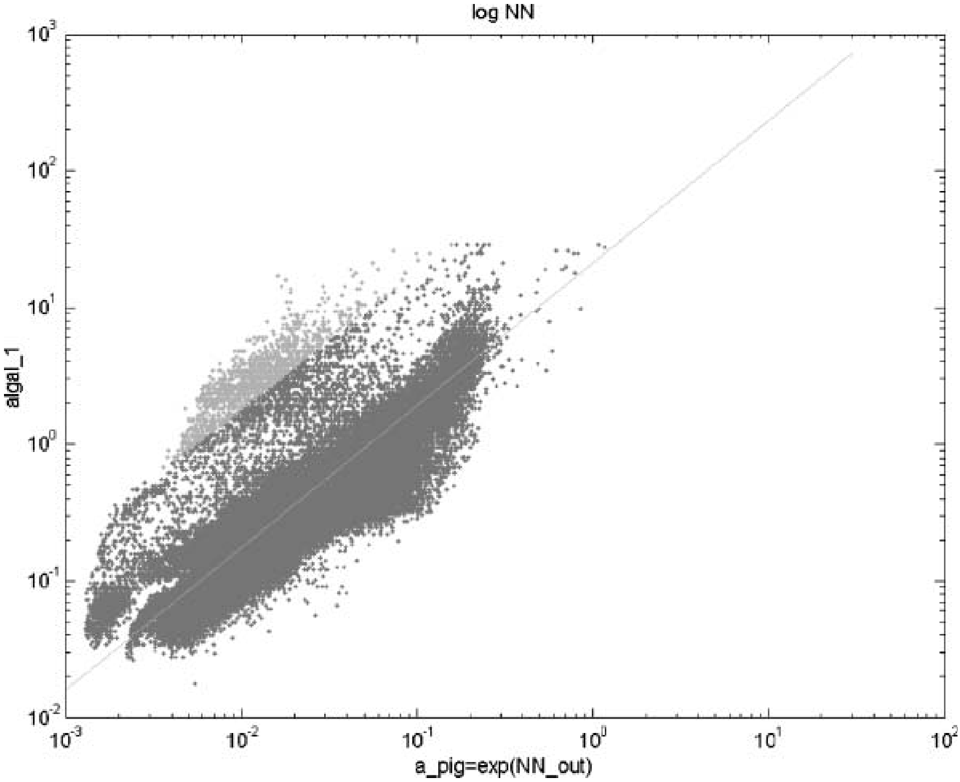

Figure 5. Relationship between pigment_absorption (output of the NN) and algal_1, the brighter dots indicate pixels at coastal case 2 water locations. The line shows the relationship algal_1=21.0a_pig^1.

the temporal difference between overflight and time of sampling in highly dynamic areas and the volume of samples compared to the volume of pixels in areas with patchy distributions on scales below pixel size. Furthermore, it is difficult to assess whether the atmospheric correction has delivered water leaving radiance reflectances with sufficient accuracy for all bands, which are then used as input to the neural network algorithm. As a consequence of these difficulties, we have to assess the performance of the algorithm by different tests and use the results as quality indicators rather than as a rigorous quality proof. Furthermore, the retrieval success, as demonstrated, may vary from pixel to pixel due to various reasons such as composition of concentrations, deviation from bio-optical model etc. Thus, it is not possible to give a general error number or range of confidence. In areas with low TSM concentrations the retrieval error for chlorophyll *a* may be in the order of 10–20% while in very turbid waters the error for a low chlorophyll *a* concentration may exceed 100%. This fact is also the reason, why we do not present error figures of tests, which might create misleading expectations.

Thus, in the following we present results from four different categories of tests in form of scatter plots and graphical comparisons: (1) test with simulated reflectances using the same bio-optical model, (2) test with simulated reflectances, which are based on an independent bio-optical model (IOCCG test), (3) comparison with alga_1 data of MERIS and (4) examples from validation cruises.

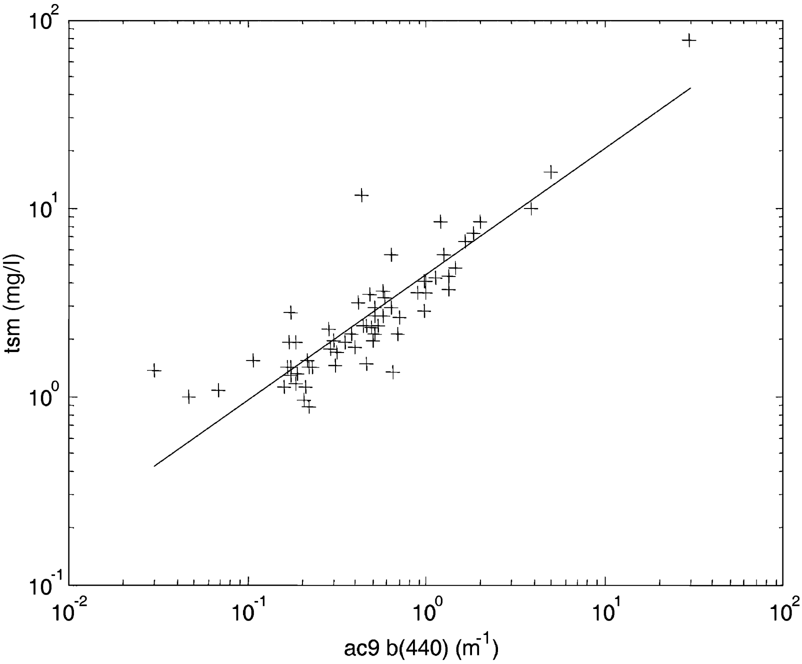

Figure 6. Relationship between total particle scattering and total suspended matter (tsm) dry weight. Data of the Southern North Sea, cruise H187.

(1) During the training process the network is always tested against a test data set, which was separated from the total set of simulated reflectances. These tests assure that the NN is not overtrained, i.e. the error resulting from the training data subset should be in balance with the error of the test subset in order to achieve a good interpolation power of the network.

Furthermore, the NN output concentrations are plotted against the concentrations of the test data set, which were used to simulate the reflectances, and which are the input to the NN. These tests can be performed for different concentration ranges. The result is an indication of the accuracy, which can be expected for different concentration ranges.

Figure 7 gives an example of such a test. In figure 7(*a*) the NN output for pigment absorption expressed in units of chlorophyll (mg m^{-3}) is tested with the full range of concentrations of *gelbstoff* and total suspended matter concentrations (table 1). It is clear that the retrieval error or uncertainty increases with decreasing pigment concentrations, because *gelbstoff* and TSM are then dominating the optical properties. If, for the same test, the concentrations of *gelbstoff* is restricted to <0.008/m and that of TSM to <2mg/l, the retrieval of chlorophyll concentrations has much less scatter also in the low concentration range (figure 7(*b*)).

(2) A second test was performed against the simulated IOCCG set of test reflectance spectra, which are based on a totally different bio-optical model than that, which has been set up for MERIS (s. IOCCG report NO. 5, Lee *et al.* (in press), www.ioccg.org/reports_ioccg.html). For this test the IOPs "total absorption" and "total scattering" of all water constituents were

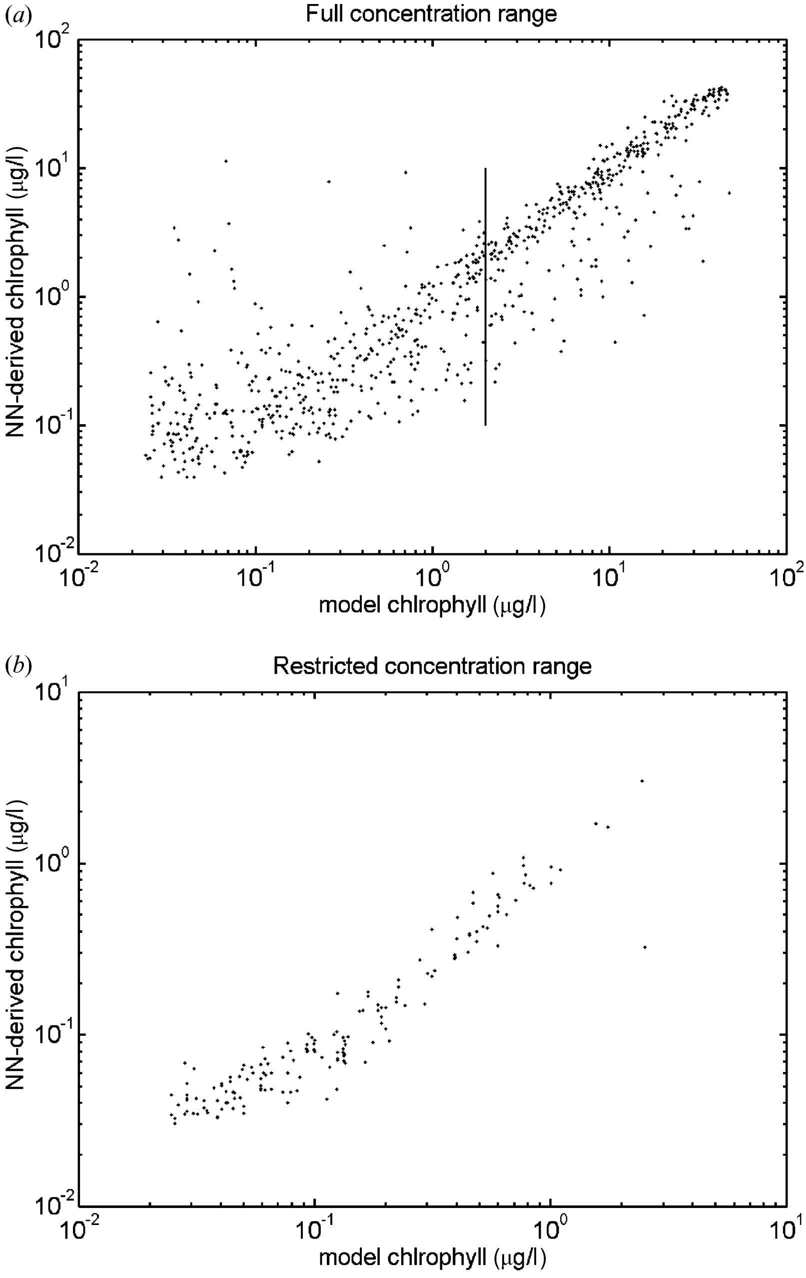

Figure 7. Points of the test set are used to compare chlorophyll *a* concentrations from the NN output with those used as input to the radiative transfer model (*a*) for the full range of concentrations of TSM and gelbstoff, and (*b*) for a restricted range (TSM<2 mg/l, *gelbstoff* absorption a(443nm)<0.008/m). The vertical line in (*a*) indicates the lower limit of a useful retrieval of chlorophyll concentration if the water contains high concentrations of *gelbstoff*.

compared. This result is presented in figures 8 and 9. From this we conclude that the NN procedure as implemented for MERIS is robust enough to cope also with conditions related to a different bio-optical model.

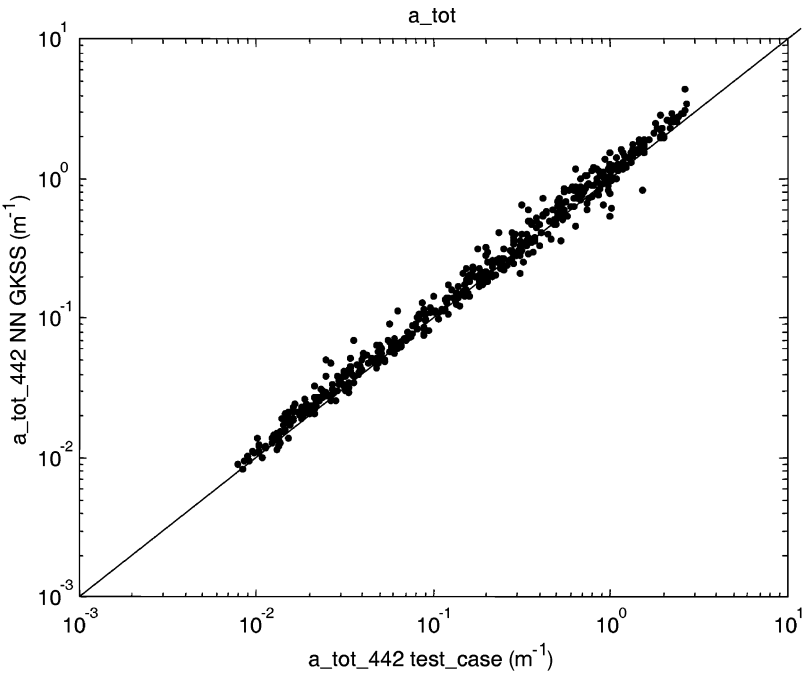

Figure 8. Comparison between the total absorption of all water constituents of the IOCCG synthetic test data set and the output of the neural net (*a*_pig+*a*_gelb combined).

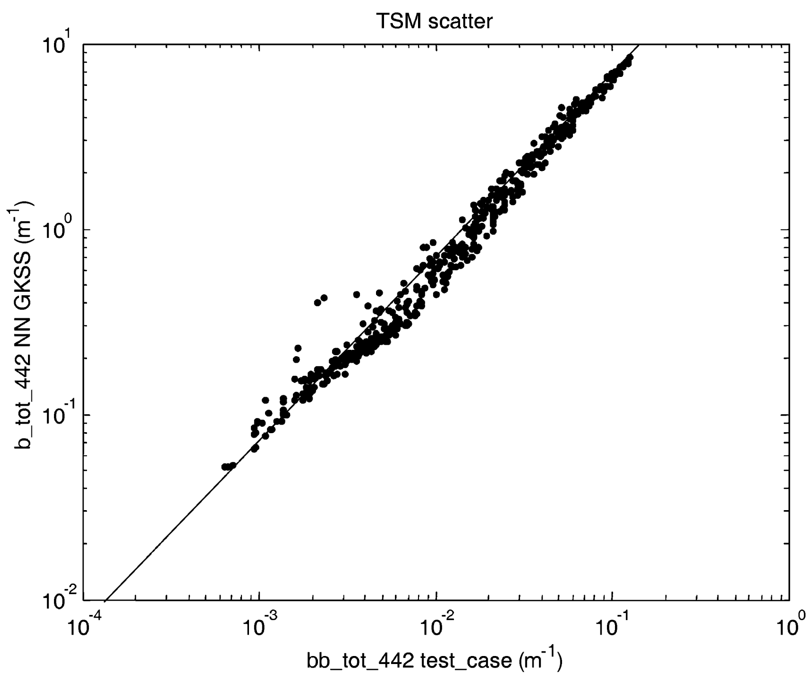

Figure 9. Comparison between the total backscattering of all water constituents of the IOCCG synthetic test data set and the output of the neural net (total tsm scattering b_tsm).

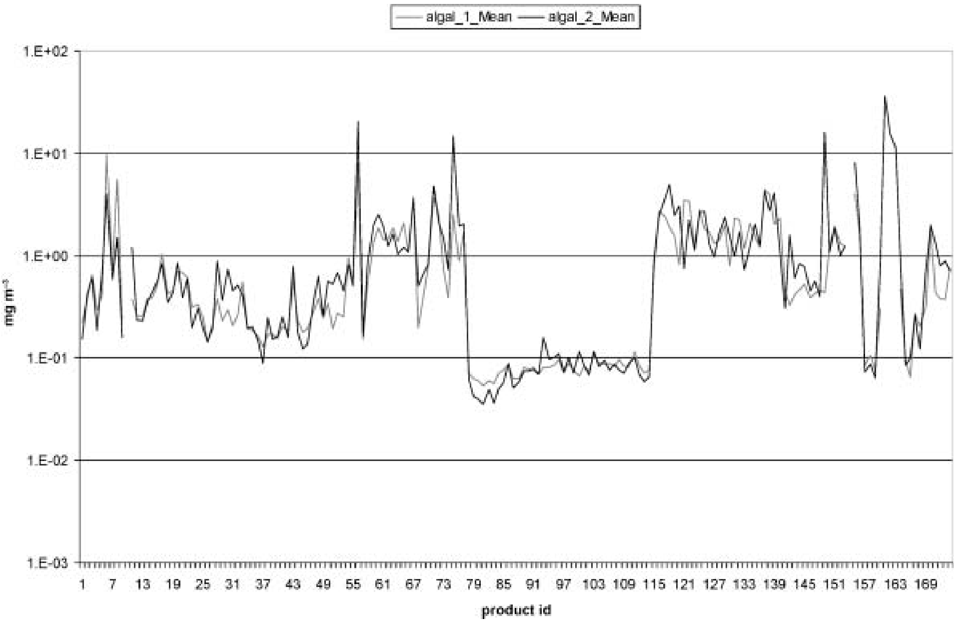

Figure 10. Mean chlorophyll concentration of water pixels without a warning flag of 170 MERIS scenes. Concentrations were produced with the MEGS 7.3 processor using the MERIS case 1 water algorithm (algal_1 and the case 2 water NN algorithm, algal_2). The scenes cover mainly different oceanic sites of the Atlantic and Pacific Ocean, the Mediterranean Sea and North Sea (courtesy of C. Brockmann, BC).

(3) For the third test the two MERIS products were compared, i.e. algal_2 chlorophyll with algal_1 chlorophyll concentrations. Both algorithms are totally independent and are based on different principles and in situ data. A set of 170 images from different oceanic regions including the North Sea, the Mediterranean Sea, the Atlantic and Pacific ocean, which cover mainly case 1 water but also case 2 water conditions, were produced with the MEGS 7.3 processor at ACRI. This set was used to compare the mean chlorophyll concentration image by image (figure 10). As one should expect clear correlations but also systematic deviations between the two algal products are evident.

(4) A number of tests were performed by comparing in situ data with the concentrations in the MERIS product. Also this test was performed not on the IOP but on the concentration level. Most of the data for the validation of MERIS are based on measurements from the ferry boat "Wappen von Hamburg" along a transect between the city of Cuxhaven at the mouth of the Elbe estuary and the island of Helgoland, which is about 50 km offshore. Along this transect the concentrations of all water constituents change due to mixing of the Elbe river water with the North Sea water. The distribution of water constituents is rather patchy and variable because of the strong tidal currents so that a direct comparison on a pixel by sample match up bases is problematic. Thus the comparison is based on the trends along this transect rather than on a direct comparison. Here only one example (figure 11) is given for the three water constituents, i.e. total suspended

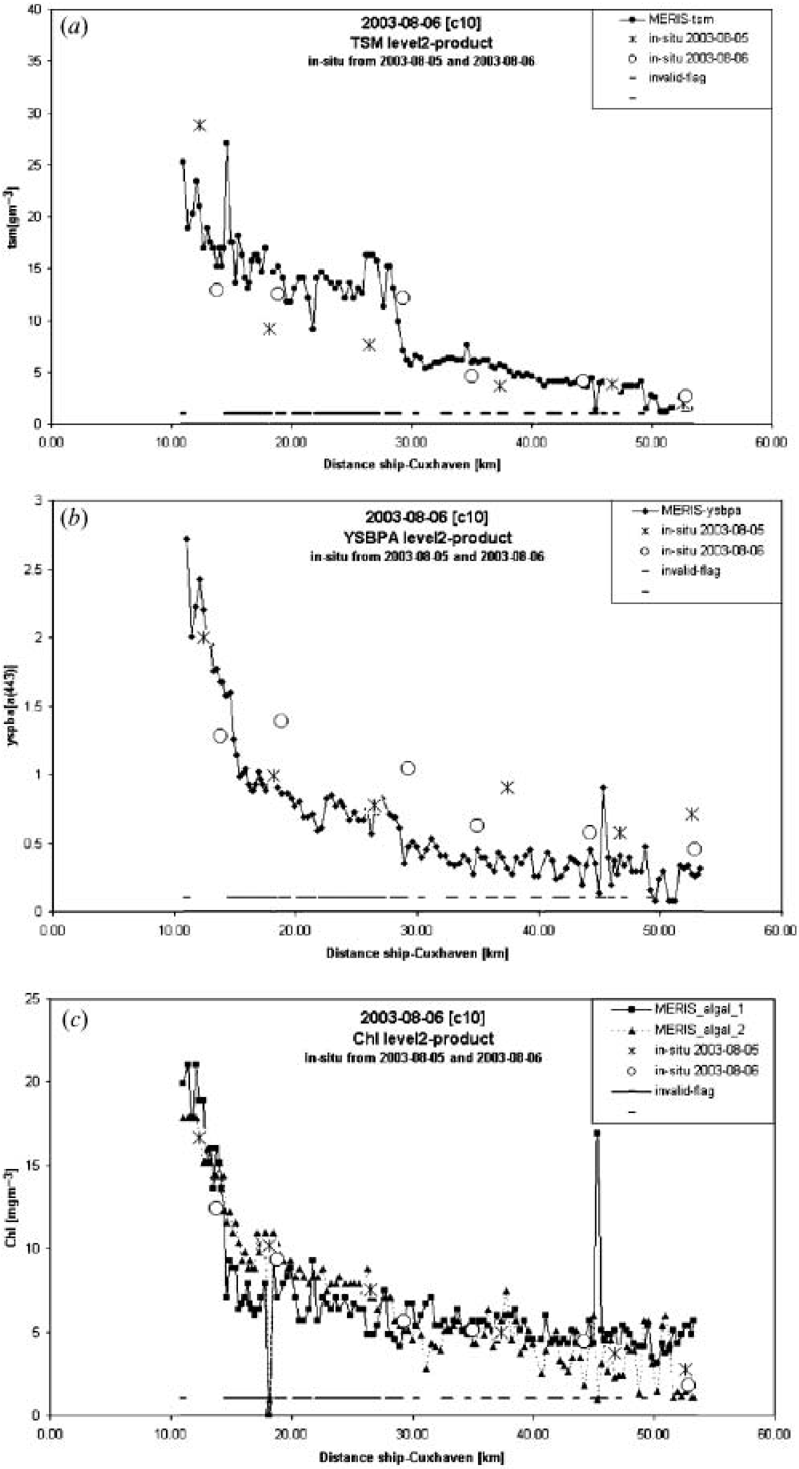

Figure 11. Concentrations of TSM (*a*), gelbstoff (*b*) and chlorophyll (*c*) derived from MERIS and in situ samples along a transect from the mouth of the river Elbe to the island of Helgoland in the German Bight (North Sea). The broken line close to the abscissa indicates pixels which are flagged 'invalid'.

matter (TSM), chlorophyll *a* and gelbstoff + absorption of bleached TSM. For information also the algal_1 product is plotted as well as the warning flags. Obvious is that the gelbstoff derived from MERIS data is significant lower than the *gelbstoff* measured in situ. The main reason for that presumably is that the atmospheric correction procedure produces a water leaving reflectance, which -with that processor version - is too high in the blue part of the spectrum.

5. Summary and conclusions

A neural network algorithm has been developed for the ground processor of the imaging spectrometer MERIS, which derives the inherent optical properties (IOPs), i.e. the absorption coefficient of phytoplankton pigments and that of gelbstoff + bleached total suspended matter as well as the scattering coeffcient of particles, all at a wavelength of 442 nm (MERIS band 2). These properties are then converted into concentrations of chloropyhll *a* and total suspended matter dry weight while the *gelbstoff* concentration is given in absorption units. The network is trained with simulated angle dependent radiance spectra so that no assumptions between radiance and irradiance reflectances have to be made as it is the case for semi-analytical irradiance reflectance models. The range of concentrations covers case 1 open ocean as well as turbid coastal case 2 waters.

Since the bio-optical model includes only standard cases the water leaving radiance reflectance of each pixel is tested if it is within the scope of the training data set. This feature is important to warn the user of out of scope spectra, which might be caused by exceptional conditions in water as e.g. caused by a Red Tide, shallow water with sea bottom reflectance or by an insufficient atmospheric correction. The threshold to switch the warning flag on is difficult to define, so that the user is asked to check this flag carefully. Unfortunately, due to restrictions of the processor software, it was not possible to include the degree of deviations between the measured and simulated spectrum to into the product.

Another important feature was to introduce a minimum reflectance threshold to avoid that low reflectances with a relative large error or even negative values would enter the network. A statistical analysis of different MERIS scenes showed that a network which was trained with a threshold of about 0.0009 sr^{-1} is robust against most uncertainties of the atmospheric correction.

The main advantage of a NN algorithm is that it provides the possibility to invert directional radiance spectra into IOPs or concentrations since it emulates the processes of a full radiative transfer code, which could not be used in an optimization loop on a pixel by pixel bases due to its high computational demand. Although the preparation of a NN is time consuming, its application in mass production is fast since it does not require any iteration.

Since 8 MERIS bands in the blue-red spectral range between 412 and 708 nm are used, the algorithm covers a wide concentration range where the most sensitive bands change from blue to red with increasing concentrations of water constituents.

The performance of the present version is indicated by the agreement between different test data sets, by the validation using match up in situ observations and by the comparison with the algal_1 product for case 1 waters. The agreements show furthermore, that the bio-optical model used to train the NN is rather universal. The separation between IOPs, which are derived from the NN inversion and concentrations, allows the user furthermore to apply his own conversion factors to the IOPs, which might be better adapted to his area of interest. However, due to the fact that all 8 bands are used, and that the atmospheric correction produces noise for spectral ranges with a low reflectance, the algal_2 product is more noisy in case 1 waters compared to algal_1, for which only 2 or 3 bands are used, which are located in the range of maximum reflectance.

The further development of the procedure, which is presently under preparation, will include the combination of the NN algorithm with an optimization procedure using the forward NN (Schiller and Doerffer 2005) as well as a confidence map, which will provide the range of uncertainty for each product on a pixel by pixel bases. Furthermore, NNs based on regional bio-optical models and models for special conditions such as exceptional plankton blooms are foreseen.

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